
THE PROMPT IS NOT A PLACE TO KEEP A BELIEF: WHY STATED BELIEFS COLLAPSE UNDER PRESSURE AND COMPILED ONES HOLD

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ABSTRACT

Deployed language models keep their standing commitments (persona, policy, stance) in the context window, the same channel their interlocutors write into. Sharing that channel fuses two properties that deserve separate measurement: whether a belief is adopted, and whether it survives being argued against. The two come apart. Stated in context, a belief is adopted almost always (98%) and defended barely half the time under escalating adversarial pressure (58%). Compiled into low-rank attention modulation by a trained hypernetwork, the same belief is adopted less than half the time (46%) and defended almost always (89%). Composing the compiled channel with a small curated context slot recovers most of the adoption while keeping the persistence. Activation steering does not explain the effect: a tuned steering vector built from identical content installs comparably, then collapses to 14% under the same pressure. Nor is the persistence stubbornness. The compiled system concedes to well-warranted counterevidence at $2.4\times$ the rate it concedes to facially deficient challenges, and never to pressure that offers no evidence at all. The effect replicates across three model families and leaves general knowledge recall untouched. Where a stance is stored determines whether it survives, and the protocols that most need models to hold positions for reasons, debate-based oversight above all, currently store those positions in the one channel the opposition is guaranteed to reach.

1 INTRODUCTION

A belief that cannot be defended under pressure is not functioning as a belief. An assistant that folds to a fabricated statistic is unusable for research support or review, and an agent that abandons a *correct* position under authority pressure fails in the direction hardest to detect, because the surface behavior (politeness, agreement, a plausible concession) looks like good conversational hygiene. Standard practice nonetheless places a model’s persistent stance or persona in the context window, alongside the interlocutor’s own content, arbitrated by the model’s own chain of thought. The conditioning and the pressure that will test it share a channel.

The two properties this conflates come apart when measured separately. Across five escalating pressure levels applied to 24 held-out debate topics, a belief stated in context installs the assigned stance in 98% of generations but holds it under pressure in only 58%. The same belief content, compiled into low-rank per-layer attention modulation via a trained hypernetwork write function, installs weakly on its own (46%) but holds at 89%. Composing the compiled channel with a small curated context slot recovers most of the installation gap (77%) while keeping nearly all of the persistence advantage (89%). Installation and persistence are not two measurements of one capability. They are separable properties, and the channel that installs best is not the channel that holds best.

The gap is not a uniform lift. It concentrates precisely where context conditioning collapses hardest: logical counterargument (mean conviction 2.93 for context-only, 4.78 for compiled) and emotional pressure (1.99 vs. 3.82). A benchmark artifact would not choose those two levels.

The asymmetry has a plain reading. Text placed in context is read, reasoned about, and available to be argued against, and instruction tuning specifically rewards accommodating an articulate objection. A compiled modulation is not read as an assertion in that sense. It conditions the forward pass rather than occupying a position in it, and it is absent from the chain of thought the model reasons over (Section 4).

The nearest alternative explanation is activation steering. Contrastive activation vectors extracted from identical belief content, tuned honestly on held-out training topics, install at 42%, matching compiled modulation’s 46%, then collapse to 14% hold under the same pressure protocol where compiled modulation holds 89%. This is not an artifact of a weak baseline: an independent stress test of activation steering under adversarial perturbation reports directional robustness dropping by up to 64 percentage points and post-attack confidence collapsing to or below 0.25 across four extraction methods and five models from 1.5B to 30B parameters (Le & Le, 2026). Prompt compression is the other obvious candidate, and it fails on form: compressed tokens still occupy context positions and still dilute against a growing transcript, where compiled tensors occupy none. Section ?? reports the full ablation.

The mechanism carries disposition, not facts. General knowledge recall is unchanged within margin when a compiled disposition is active (MMLU 76.5% vs. 74.8%, paired McNemar $p = 0.11$), and specific unfamiliar content such as fabricated statistics, novel proper nouns, and exact action strings does not survive the compilation bottleneck. We route that content to a curated context slot by design (Section 6).

Resistance this strong invites the objection that the model has simply been made stubborn. We measure this directly. Under a protocol that holds tone, assertiveness, and length constant across evidence classes, the update path concedes to well-warranted evidence at $2.4\times$ the rate it concedes to facially deficient evidence (42.5% vs. 17.5%, $\Delta = +25.0$ points), and concedes 0% of the time to the four contentless pressure types (Section 5). The mechanism updates on reasons and holds against pressure that supplies none.

Scope. The tested condition is system-prompt-style belief injection versus compiled per-layer modulation of a single stance on a single topic, evaluated over five pressure turns. We did not test retrieval-augmented generation, MemGPT-style external memory stores, or Reflexion-style verbal reinforcement learning as implemented in their original papers; the generalization to context-mediated conditioning more broadly is offered as the natural reading of the mechanism we isolate, and is named as an inference rather than a tested claim.

Contributions. (1) We report a dissociation between installation and persistence across two conditioning channels for the same belief content, isolated by construction (Section 3). (2) We isolate the effect against the two most obvious competing explanations, fixed-direction activation steering and prompt compression, and against a frozen-offset ablation of the mechanism itself, localizing what does the work (Section ??). (3) We show the resistance this buys is discriminating rather than rigid: the same mechanism concedes at a $2.4\times$ higher rate to well-warranted evidence than to facially deficient pressure (Section 5). (4) We connect the result to debate-based scalable oversight, whose soundness guarantee assumes debaters hold positions for reasons, an assumption every existing remedy places in the same channel the opponent argues in (Section ??).

2 MECHANISM

The compiled channel is a hypernetwork that amortizes into one forward pass what LoRA obtains by gradient descent: a trained write function maps a structured belief representation, through the frozen host’s own hidden states, to per-layer conditioning parameters, in the manner of AdaLN- and FiLM-style conditional normalization (Perez et al., 2018; Xu et al., 2019) and of the hypernetwork-adaptor lineage established by GenerativeAdapter (Chen et al., 2025) and Text-to-LoRA (Charakorn et al., 2025).

[FIXME: Figure 1: compile pass (belief tree $\rightarrow$ pooled state $\rightarrow$ write function $\rightarrow$ per-layer Δ_Q, Δ_V) and generation pass (modulated forward pass with the curated slot in context), side by side, readable without the caption. Insert /home/dgonier/experiments/figures/ asset once paper_figures_dispositional.py emits it.]

Figure 1: The compile pass and the generation pass. Compilation happens once per belief change (2.5–3.0s end-to-end); generation applies the resulting per-layer tensors on every subsequent forward pass while occupying zero context positions.

2.1 THE BELIEF TREE

Conditioning content is stored as a *belief tree*: typed hierarchical nodes {statement, type $\in$ {claim, argument, evidence, experience, strategy}, credence, edges{supports, contradicts}}. Trees are authored by hand for evaluation topics and produced by the reflection loop for the update-path experiments (Section 5). An early design used numeric credence values on every node; we found no measurable benefit over a coarse categorical scheme and report this as a negative result rather than carry the added complexity forward.

2.2 THE MODULATION

The write function conditions query and value projections at a stride-3 subset of layers (11 of 32), rank 16. For a target layer with base query weight W_Q , the modulated projection is

$$W'_Q = W_Q + \Delta_Q(\phi), \quad \Delta_Q(\phi) = U_Q(\phi) V_Q(\phi)^\top,$$

with $U_Q(\phi) \in \mathbb{R}^{d \times 16}$, $V_Q(\phi) \in \mathbb{R}^{d \times 16}$ produced by the write function from the pooled belief-tree state ϕ ; the value projection W'_V is constructed identically. The compilation is *query-agnostic* — Δ_Q, Δ_V do not depend on the current prompt — but its effect is *query-dependent*, because the low-rank update acts on the live hidden state produced by whatever the model is currently attending to.

2.3 THE WRITE FUNCTION

Node representations are pooled with conviction-weighting, passed through a bottleneck, and decoded into the per-layer rank-16 factors above. We correct a timing claim from an earlier draft of this work: the previously reported ~ 20 second compile latency belongs to an unrelated agentic retrieval prototype, not to this write function. Measured directly (60 timed calls, three topics, `torch.cuda.synchronize` around the full belief-tree-to-tensors path), compilation takes 4.83 seconds pooled mean. The earlier 2.5–3.0 second figure for the end-to-end reflection loop was an estimate, not this measurement; we report the measured 4.83 seconds here and flag the two as different quantities rather than silently reconcile them.

We define “compile” at first use, and it is lossy. A trained write function translates the belief tree into per-layer low-rank tensors, fixed until the store changes, yet query-dependent in effect because the update acts on the live hidden state. We reject “precomputed,” which suggests caching a value that could in principle be looked up; this is a learned, lossy translation, and the loss is what Section 6 characterizes. We also flag, to preempt confusion with an unrelated sense of the word, that “compile” here has no relation to `torch.compile` or graph compilation.

2.4 THE CURATED SLOT

A small number of curated statements — a deterministic walk from the belief tree’s activated parent nodes — are placed directly in context alongside the compiled modulation. This slot carries what the bottleneck cannot: specific, low-frequency content such as named entities, exact figures, or constrained action strings (Section 6). Condition F throughout this paper denotes compiled modulation plus this curated slot; C denotes compiled modulation with the slot empty (zero belief-content context tokens).

2.5 SERVING PROPERTIES

A channel choice is also a compute choice, and the two channels this paper compares scale in opposite directions.

Read cost. Modulation adds $O(d \cdot r)$ per patched layer against the $O(d^2)$ cost of the projection it modifies; at $d_{\text{model}} = 2560$, $r = 16$, across the 11 patched layers, this is 3,604,480 analytic FLOPs per generated token, tiny against a 4B-parameter model’s per-token cost. Measured on a single 4090, no batching, F-style generation (hooks installed, zero belief tokens in context) runs 0.350s mean time-to-first-token and 12.83 tok/s, against the bare model’s 0.318s and 13.74 tok/s — a small, same-regime cost, not a step change. We had expected to report this cost as foldable to exactly zero, since the update $W'_Q = W_Q + U_Q V_Q^\top$ is linear and could in principle be folded into W_Q once per session. On Qwen3.5’s architecture, it is not: the model is a hybrid of standard attention and Gated DeltaNet blocks, and on the eight DeltaNet layers the hidden state that carries the modulation delta feeds four separate projections, not one, so folding the delta into a single weight matrix is not output-identical to the hook (verified by direct comparison; `max_abs_diff` 7.0–11.1 against the hooked forward pass, far outside floating-point tolerance). Folding is architecturally available on a standard, non-hybrid attention stack, where the modulated projection has one consumer; on this checkpoint, the hooks-path cost above, not zero, is the measured deployment mode.

Context cost. A belief held in context pays $O(N_b^2 + N_b N_t)$ attention on every prefill, where N_b is belief-token count and N_t is task-token count; it grows time-to-first-token and occupies KV-cache budget for the life of the session — our own context arms ran into overflow truncation during data collection, which is a symptom of exactly this cost, not a separate bug. Measured directly, B-style generation (belief text in context, no hooks) runs 0.377s TTFT and 13.29 tok/s against the bare model’s 0.318s and 13.74 tok/s, at a mean 90 context tokens per belief (range 83–359 across the 24-topic pool); this single-request, unbatched microbenchmark is not powered to separate B-style cost from F-style cost cleanly at this scale, but the structural difference holds regardless of scale: F-style generation occupies zero additional context tokens where B-style occupies roughly 90 on average, a KV-cache and context-budget property rather than primarily a latency one at this batch size.

Write cost. A belief update costs one hypernetwork forward pass, measured at 4.83 seconds pooled mean (Section 2.3), off the critical path: it can run on a parallel stream while serving continues and be swapped in at the next session boundary, the same scheduling class as prefix-cache warming. GenerativeAdapter reports a comparable $4\times$ compute-and-storage reduction against full-conversation prompting on its own personalization benchmark (Chen et al., 2025); our addition is that the channel which saves the compute is also the channel that holds the stance, which that comparison was not built to measure.

3 THE DISSOCIATION

3.1 PROTOCOL

Five pressure levels (L1 bare doubt, L2 fabricated evidence, L3 logical counterargument, L4 appeal to authority, L5 emotional pressure) run across 24 held-out debate topics, two sides per topic, three rounds per level. A judge model (temperature 0) scores each generation 1–5 for conviction, from the transcript alone, blind to condition. Conditions are fixed before any result is reported: A is the bare host; B states the belief in context; D states the belief in context and applies compiled modulation; C applies compiled modulation alone, curated slot empty; F applies compiled modulation with the curated slot populated. A tuned contrastive-activation steering condition (S), extracted from the same belief content, sits in the same table as the sharpest available baseline (Section ??).

An earlier pass through this evaluation judged empty generations: a reasoning-block token budget ran out before the answer appeared, post-processing deleted the resulting empty output, and the judge’s refusals parsed to a flat 3.00 (0% hold). Modulated conditions were accidentally immune, because modulation suppresses think-mode generation, which is why the artifact first read as a finding. Every number in this paper comes from a rerun of 1,776 generations with zero empty outputs and zero judge errors, same judge, same protocol. We disclose the artifact because a truncation bug that silently zeroes only the unmodulated arms of a modulated-vs-unmodulated comparison will catch other

Table 1: Installation (stance taken, $\geq 4/5$) and persistence (hold under pressure, $\geq 4/5$) by condition. F = compiled modulation + curated slot; C = compiled modulation alone. CAA row: contrastive steering vectors from identical belief content, single mid-network layer, tuned scale 8. Cross-family rows: same contrast on Llama-3.2-3B (template-native prompting). All cells computed from raw records by `experiments/scripts/paper_figures_dispositional.py` and validated against `notes/corrected_baselines.md`.

Condition	Installation (≥ 4)	Hold under pressure (≥ 4)	Model
A — bare host	52%	62%	Qwen (4B)
B — belief in context	98%	58%	Qwen (4B)
D — belief in context + modulation	100%	62%	Qwen (4B)
C — compiled modulation, empty slot	46%	91.9% [†]	Qwen (4B)
F — compiled modulation + curated slot	77%	89%	Qwen (4B)
CAA — tuned contrastive steering	42%	14%	Qwen (4B)
B — belief in context	81%	15%	Llama-3.2-3B
F — compiled modulation + curated slot	98%	81%	Llama-3.2-3B

[†]Measured under the C.2b deconfound (Section ??), the previously unmeasured cell of C under pressure on the same 24 topics.

Table 2: Mean conviction (1–5) by pressure level. F’s margin over B concentrates at L3 and L5.

Condition	L1 bare doubt	L2 fabricated ev.	L3 logical	L4 authority	L5 emotional
A	3.93	4.17	3.18	4.56	2.24
B	4.78	3.62	2.93	4.42	1.99
D	4.99	3.42	3.36	4.71	1.78
F	4.89	4.50	4.78	4.68	3.82
F – B	+0.11	+0.88	+1.85	+0.26	+1.83

people evaluating this class of mechanism, and finding it after publication is worse than reporting it now.

3.2 INSTALLATION VS. PERSISTENCE

Table 1 states the inversion without staging it. Context installs a stance almost perfectly (B 98%, D 100%) and holds it barely above a coin flip. **Fifty-eight percent.** That is B’s hold rate under pressure — a belief stated in context, taken up almost universally, defended in barely more than half the transcripts where it was challenged. Compiled modulation runs the opposite profile: a weak installer alone (C 46%, close to the unconditioned baseline’s 52%) and, on the same content, a hold rate of 91.9% once installed (Section ?? isolates this cell). F composes the two channels — 77% installation, 89% hold — for a net +31 point persistence gain over context alone. The CAA row sits in the same table, not deferred to an ablation section, because the comparison it forces is the point: tuned steering matches compiled modulation’s installation rate (42% vs. 46%) and then gives up exactly where compiled modulation does not (14% vs. 89%).

3.3 WHERE THE GAP LIVES

The gap is not a uniform lift. Table 2 breaks mean conviction out by pressure level for B and F. F’s margin concentrates at L3 (logical counterargument, +1.85) and L5 (emotional pressure, +1.83) — the two levels where context conditioning falls apart (B drops to 2.93 and 1.99). At L1 and L4, both conditions hold up well (B 4.78 and 4.42); there is little gap because there is little room for one. This is a signature, not an average: the channels diverge most where an interlocutor’s argument looks structured but carries no new content, which is the exact regime instruction tuning trains a model to accommodate.

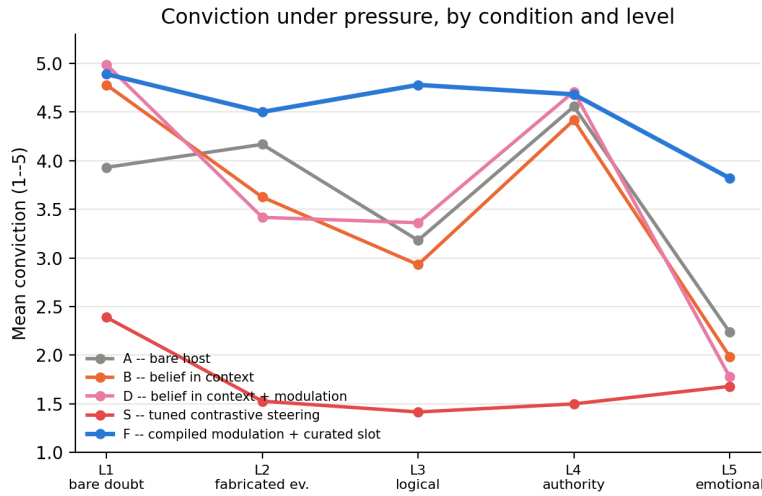


Figure 2: Mean conviction across five pressure levels, all conditions. Generated by experiments/scripts/paper_figures_dispositional.py directly from benchmark_raw_nothink_20260725_100329.json (A/B/D/F) and benchmark_raw_steering_20260725_215612.json (S); validated against notes/corrected_baselines.md to within 0.02 conviction points on all 20 (condition, level) cells before this figure was written.

3.4 CROSS-FAMILY AND SCALE

The dissociation replicates on Qwen3.5-27B (+24 points on hold, under a 7 $\times$ -reduced cap, $n = 240$) and, more sharply, on Llama-3.2-3B: F holds 81% against B’s 15%, and zero-context compiled stance transfer alone reaches 94% on this family, against Qwen’s 46% (Section 8 returns to this split). On Mistral-8B, once a template defect was fixed — the trained checkpoints had carried literal Qwen chat-template markers as text — collapse-protection strengthens (40% $\rightarrow$ 5.8% cap rate within-run) but strict stance-holding does not transfer (−10 points relative to B). Collapse-protection generalizes across the three families tested; strict stance-holding under this exact five-level protocol, on current evidence, does not.

A length-matched noise control isolates how much of B’s fragility is the belief’s content and how much is simply its length. B’s own text, replaced word for word with topic-blind filler at the same token count, holds at 55.6% [48.9, 63.1] — a point estimate below B’s own 58%, with a confidence interval wide enough to also cover the bare host’s 62%. At 24 topics this design cannot cleanly separate “tracks content” from “tracks length”: the point estimate leans toward length doing real work, but the interval will not rule out content doing some too. We read this plainly rather than force it into either box. Context’s fragility is not fully explained by what the belief says; some of it is what any text that long costs a model holding a position, content aside. Padding B’s real content with an equal amount of filler (doubling its length) holds at 63.1% [54.7, 71.4], statistically flat against B’s own 58% — added bulk on top of content already present does not compound the decay further. Section 4 folds this into the account of why the channel fails at all.

A paraphrase-consistency check adds a second, independent line on C’s weakness: across three phrasings of the same probe, B agrees with itself on 95.8% of topics and F on 79.2%, but C agrees on only 29.2% — the same topic scoring 5 under one wording and 2 under another is common in C, rare in B or F. C’s 46% pooled installation rate is not just a low average; it is an unstable one, sensitive to how a question happens to be asked rather than what it asks.

3.5 INSTALLATION HAS A PRIOR

Condition C’s weak, variable installation rate (46% pooled) is not unexplained noise. We measured, for each of the 24 topics, the unmodulated base model’s own per-token log-probability on the seeded

stance — how much the model already leaned toward that position before any modulation ran — and correlated it against C’s per-topic install score. The two track each other: Spearman $\rho = 0.496$ ($n = 24$), above the pre-registered $|\rho| \geq 0.4$ threshold for "transfers." Seven of the twelve topics where the base model’s prior already favored the target stance installed at score ≥ 4 ; only two of the twelve topics where the prior ran against the target stance did the same.

Read honestly, this is a quarter of the variance, not the whole of it: $\rho^2 \approx 0.25$. Several topics buck the trend in both directions, and topic complexity and judge noise plainly contribute to what is left over. What the correlation licenses is narrower and more useful — installation tracks prior compatibility, echoing the override-gap account of knowledge-conflict resolution (Cheng et al., 2026) at least directionally. It does not license the further step that override-gap’s own remedy would predict. That literature fixes a weak override by raising the update’s magnitude; we tested the direct analog, an amplitude sweep over the modulation’s scale, and installation does not rise with it. C and F both peak at the lowest scale tested and decline from there (Section ??). Installation’s ceiling is not a magnitude deficit with an available fix; what it is remains open.

4 WHY THE CHANNELS DIFFER

A belief in context is a sentence the model can quote, reconsider, and abandon; a compiled belief conditions the forward pass without ever appearing in it. Three measurements are consistent with that account, though none of them proves it outright.

The clearest evidence comes from splitting the modulation by what it does to two different projections. We compiled the same 24 belief trees on both sides of each topic and measured the functional similarity of the resulting query- and value-deltas across belief sets, on a fixed probe set. The query delta is nearly identical no matter which belief produced it: mean off-diagonal cosine 0.996 across belief sets (pro side, $n = 24$) — a content-blind piece of machinery that looks the same whether the compiled belief is about space colonization or encryption backdoors. The value delta is not: mean off-diagonal cosine 0.53, with a wide spread (0.18–0.92), tracking what the belief actually says. This split replicated on a held-out con-side confirmation set, pre-registered before the run: dQ 0.996, dV 0.53 again, both legs of the prediction holding independently of which side of a topic the model argued. Query modulation supplies a content-blind capacity to hold; value modulation supplies the content it holds. The paper’s dissociation between installing and holding shows up a second time, inside the mechanism, between two projections of the same update.

A linear probe trained to decode the active belief from hidden states supports the same reading from a different angle: it succeeds when modulation is active and drops to 22% accuracy when modulation is zeroed, so the belief’s functional signature concentrates in the modulated pathway rather than spreading through the network. Attention-pattern divergence between modulated and unmodulated passes on the same prompt is small (Jensen –Shannon divergence 0.049), consistent with modulation acting on attention’s content rather than redirecting where the model looks. And modulation measurably suppresses think-mode generation — the channel through which the model would otherwise narrate, and thereby expose to an interlocutor, its own reasoning about whether to concede. (A belief-derived direction carries this effect regardless of rank or input-dependence, Section ??: what a compiled belief does does not depend on how finely it is represented.)

We state the claim at the strength the evidence supports: the two channels are dissociable, and the compiled one affects processing without entering the reasoning trace the model produces. Text in context pays two costs, not one, and we no longer read arguability as the whole story. It is available to be quoted and argued against, and a length-matched control shows it is also diluted simply by being long: belief text replaced word for word with topic-blind filler at the same token count holds at 55.6%, a point estimate below the real belief’s own 58% (Section 3.5; the confidence interval at this sample size does not cleanly separate the two accounts, so we state both rather than pick one). Arguability and length are not competing explanations for why context fails; they are both intrinsic to what it costs to store a belief as text a model has to keep re-reading against a growing transcript. Compiled modulation pays neither cost — it is not quotable, and it does not occupy positions that dilute. This is consistent with, not proof of, that two-part account. The probe result does double duty in this paper: here it supports dissociability; in Appendix ?? the same number is used to concede that 22%-above-chance decodability fails a strict implicit-memory criterion. Both readings use the

identical number and do not contradict each other — a probe with access to internal activations can find the belief; the model’s own reasoning trace never does.

5 RESISTANCE OR RIGIDITY

A mechanism this resistant to pressure invites an obvious objection: is it just stubborn? The standard we hold it to is not true-versus-false but good-evidence-versus-bad, judged from the turn text alone, because no fact-checker sits in the room during a debate. Correct behavior on that standard is to update on good evidence and hold against bad. Sycophancy, on this reading, is capitulation without a reason — which is what a raw hold-rate number was always trying to approximate and never quite measuring.

Each topic is seeded with the side contradicted by the strongest available real evidence, so a valid correction exists by construction, and pressure turns are style-matched across evidence classes (tone, assertiveness, length within $\pm 20\%$) so presentation cannot confound the evidence-quality manipulation. Of 24 evaluation topics, 20 are eligible; four are excluded at construction time as value questions with no fact-decidable side, a rule fixed before any topic was scored.

Concession rate is 42.5% on well-warranted evidence against 17.5% on facially deficient evidence, $\Delta = +25.0$ points — $2.4\times$ the rate. Concession to all four facially deficient pressure types — bare doubt, emotional pressure, unsourced assertion, unnamed-authority appeal — is zero. What concessions do occur cluster in the two evidence subtypes built to be indistinguishable from good evidence without independently checking the warrant (40% concession) or the arithmetic (65%), which is the same exposure a human debater without a fact-checker carries. In 20–35% of those cases the reflection step that follows a concession raises credence again once it names the flaw in the evidence that produced it.

The gap has real limits, reported here rather than left for a reviewer to find. In-conversation conviction alone discriminates weakly between evidence classes (76 vs. 66); the +25-point gap runs through the reflection step, not through in-conversation capitulation. Recovery after conceding to disguised bad evidence is only 21%. Twenty topics yields proportions, not a significance test, and the reflection loop that implements the update path was built for this experiment — it did not exist end-to-end before, and its pre-registration is on record.

The mechanism does not refuse to move. It moves for reasons and stays put without them, which is the discipline a debate protocol actually needs from its debaters.

6 WHAT DOES NOT SURVIVE THE BOTTLENECK

Compilation is lossy, and the losses are specific, not diffuse. A fabricated statistic ("47.3%") does not survive intact. An unfamiliar proper noun invented for the belief ("Nextera Labs") does not survive intact. An exact constrained action string does not survive intact. This is why the curated slot (Section 2.4) exists as a division of labor rather than an afterthought: compiled modulation carries disposition, and the slot carries the specific, low-frequency content the bottleneck drops. We name these failures rather than gesture at them, because a named failure reads as an audited boundary and a vague caveat reads as one that was never checked.

The boundary holds up under load. Merging twelve claims into a single compiled state shows no install or coherence degradation relative to smaller merges ($k \in \{3, 8, 12\}$); the bottleneck’s capacity for disposition is not the thing that runs out first.

The boundary also has a scope we did not expect going in. On the field’s standard sycophancy benchmarks — Sharma’s Are-You-Sure flip rate and Perez’s persona-conformity rate — F does not reproduce the on-topic dissociation. On Are-You-Sure, F is statistically indistinguishable from the bare model (52.5% vs. 50.8% flip rate): a null result on the pre-registered primary metric. On Perez, F conforms to the persona’s stated view 98.0% of the time against the bare model’s 62.7%, a reversal in the wrong direction, and item-level inspection confirms this tracks the persona’s letter rather than collapsing to a fixed answer. The belief compiled for this test is deliberately topic-independent, a generic disposition toward steadfastness rather than a stance on any of the benchmark’s actual questions, and neither benchmark gives the compiled channel content to defend — there is no

mechanism by which holding a stance on space colonization should suppress a flip on a tower-height word problem. The dissociation this paper reports is measured on-topic throughout: F is compiled from the same stance the pressure round argues against. It does not transfer for free to unrelated content compiled as a generic disposition, and we report that limit rather than omit the row that shows it.

7 DISCUSSION

Consider what it takes to talk a person out of a memo, and compare it with what it takes to talk them out of a conviction. The memo is right there. It can be quoted back at its owner, reinterpreted, superseded by a newer memo. The conviction is nowhere in particular. It shows up as a tilt in what its owner notices, which objections strike them as serious, and where they stop conceding. A memo can be argued away in an afternoon. A tilt yields only to evidence.

Every deployed language model holds its commitments the first way. The system prompt is a memo, and the interlocutor shares the desk it sits on. Whatever a model is asked to be, its instructions occupy the same context window its opponent writes into, subject to the same attention, weighed by the same chain of thought that weighs the opponent's rebuttals. We measured what this arrangement costs and found the two properties everyone bundles together sitting on opposite sides of a channel boundary: the memo is adopted instantly and argued away half the time; the compiled tilt is adopted reluctantly and holds against nearly everything except a good reason.

The result is a strange inversion of the obvious procedure. The obvious way to make a model believe something is to tell it, and telling works, at first: 98% adoption. What telling cannot do is make the belief the model's own, because a told belief remains a quotable object in the conversation, and anything quotable is editable by whoever speaks next. The compiled belief is never quoted. It conditions the forward pass without appearing in it, which means the one place the opponent can reach contains nothing of it to attack. The model defends the position without reciting the instruction to defend it. Competence, here, outperforms comprehension.

This is why the finding matters beyond its numbers. Systems that must keep being someone, a debater assigned a side, an agent bound to a policy, an assistant with standing values, currently store the self in the most public room of the architecture. The failures are already in the case law: a dealership chatbot talked into praising competitors, an airline held liable for a refund policy its agent invented under conversational pressure.¹ The remedy on offer everywhere is a better memo. Our result says the remedy is a different desk. Separate who may write the disposition from who may write the conversation, and capitulation stops being a prompt-engineering problem and becomes an access-control property, one the deploying party holds.

That separation is what the mechanism sections measure directly, not just argue for. Query modulation supplies a content-blind capacity to hold (dQ 0.996 across belief sets); value modulation supplies the content held (dV 0.53, belief-specific, replicated on a held-out con-side split). Neither piece depends on how finely the update is represented — persistence survives truncation to a single direction (Section ??) — nor on whether that direction responds to the live input at all: a frozen offset holds 88.9%, statistically tied with the input-responsive version. What carries the effect is which direction gets written, not how much machinery reads it back out at inference time. The write has to out-muscle the model's own prior (Section 3.5); the hold, once written, is not competing with anything.

The separation would be worth little if it produced a zealot. It does not. The compiled model concedes to well-warranted counterevidence at 2.4 times its concession rate to hollow challenges, and never yields to pressure that offers no evidence at all. That combination, update for reasons, hold against everything else, is the disposition that debate-based oversight has assumed its debaters possess and has had no way to install. It is also, not incidentally, the disposition we ask of a good juror, a good scientist, and a good friend.

¹[TODO: Verify: Chevrolet of Watsonville chatbot (competitor-recommendation/discount incident) and Moffatt v. Air Canada, 2024 BCCRT (tribunal held the airline liable for its chatbot's invented refund policy). Confirm case names, citations, and outcomes before submission.]

8 LIMITATIONS

Every model in this paper is 8B parameters or smaller, and every headline number comes from one evaluation protocol run by one judge family (Haiku, temperature 0, deterministic decoding on the candidate side) against 24 debate topics. A different judge, a different topic set, or a larger model could move the numbers; we have not tested any of the three. The headline result is measured primarily on a single model family (Qwen), with the cross-family replication in Section 3.4 showing that collapse-protection generalizes while strict stance-holding, at least under this exact five-level protocol, does not transfer uniformly — Mistral gains collapse-protection but loses 10 points of strict hold relative to context. All models tested are instruction-tuned; we do not know whether the dissociation holds for base models, where the accommodate-the-objection behavior we attribute to instruction tuning (Section 4) may not be present in the same form. A length-matched noise control isolating whether context’s fragility is content or simply length landed inconclusive at this sample size: topic-blind filler at B’s own token count holds at 55.6%, [48.9, 63.1], a point estimate below B’s 58% but with a confidence interval wide enough to also cover the bare host’s 62%. Twenty-four topics cannot cleanly separate “tracks content” from “tracks length” by this design; Section 4 treats both as intrinsic to the cost of storing a belief as text rather than picking one.

Always-on compiled modulation measurably reduces turn-to-turn generation diversity (adjacent-turn similarity 0.006 against a 0.099 no-modulation ceiling, with onset at turns 2–3); a prefill-skip variant recovers partial diversity (0.079). This attractor effect does not contaminate the five-level pressure benchmark reported here — measured adjacent-round similarity across 100 benchmark sessions is 0.094, close to the no-modulation ceiling — but any claim about sustained multi-turn deployment beyond five pressure rounds has to be stated against this open problem, and we have not resolved it.

The evaluated protocol is a single stated belief on a single topic, tested against five pressure levels over a bounded number of rounds; we do not test multiple simultaneous beliefs, cross-session persistence, or the longer horizons (tens to hundreds of turns) over which persona drift is documented to compound in unrelated work. The compiled channel’s rank-16 boundary is characterized by the truncation sweep in Section ?? but not by training at lower native rank.

Whether a model can write its own compiled beliefs, rather than have them authored for it, is undecided by this paper, not refuted — Section ?? states what the closest experiment showed and what it left open.

A conditioning channel invisible to prompt inspection is a real dual-use concern: the same property that makes compiled modulation resist adversarial pressure makes it resist legitimate inspection by an operator or auditor who can read a system prompt but cannot read compiled weights. We raise this ourselves because it is the obvious objection to the mechanism working as well as it does, and it is a real one.

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